

BIO 4AE3 - 10 Nov 2025

- MALARIA control.
- money, communications, politics.
 - ↳ antimalarial drugs.
 - vaccine.
 - children.
 - eff against symptoms

VECTOR control.

- indoor residual spraying -
- treated bednets -
- biocontrol?

Gambusia - 'mosquito fish'

- improving housing.

malaria resistance to antimalarials.

• quinine.

chloroquine

artemisinin, combination therapy

- try to eliminate falciparum malaria from SE Asia.

OR prevent spread?

INSECTICIDE resistance

DDT! side effects

evolution of resistance

[de novo mutation?]

• other methods -

- sterile male release

irradiation; Wohlbachia.

• gene drive -

- shift sex ratios toward males.
- reduce vector competence

- evolution - resistant insecticides



late-acting mutations



• MEDAWAR.

• DRIFT.

• PLEIOTROPY.

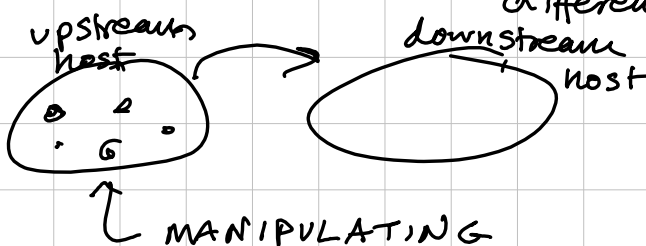
• LATE-ACTING insecticides.

HOST MANIPULATION.

parasites "want" to increase transmission.

- high fecundity (but maybe this increases virulence?)
- increase probability of transmission.

SIGNALING - getting hosts to behave differently.



Dicrocoelium dendriticum

gut fluke.

cows → ants

definitive host (sexual reprod)

modifies ant behaviour.

~ crawl up grass stems and die.

"phylogenetically old" sleeping behaviour

↳ KIN selection.